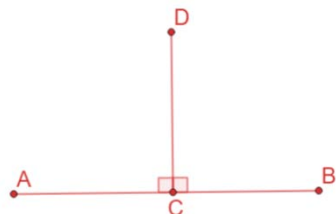


Geometry
Unit 2
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

In questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a diagram using the given information for each representation.

1. If two supplementary angles are congruent, then they are right angles.



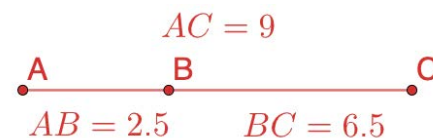
Theorem
(Sketch answers may vary)

2. A midpoint divides a segment into two congruent parts.



Definition
(Sketch answers may vary)

3. If a point, B , is drawn on a line segment AC , then AC is the sum of AB and BC .



Postulate
(Sketch answers may vary)

Complete the table.

	Property	Equality	Congruence
4.	Transitive	If $m\angle ABC = m\angle DEF$ and $m\angle DEF = 45^\circ$, then $m\angle ABC = 45^\circ$	If $\angle ABC \cong \angle DEF$ and $\angle DEF \cong \angle GHI$, then $\angle ABC \cong \angle GHI$
5.	Reflexive	$m\angle JKL = m\angle JKL$	$\angle JKL \cong \angle JKL$
6.	Symmetric	If $m\angle XYZ = 81^\circ$, then <u>$81^\circ = m\angle XYZ$</u>	If $\angle XYZ \cong \angle FGH$, then $\angle FGH \cong \angle XYZ$

7. Write the conclusion for the statement.

If $m\angle JKL + m\angle SMW = m\angle XYZ + m\angle DNF$ and $m\angle XYZ = m\angle SMW$, then...
 $m\angle JKL + m\angle SMW = m\angle SMW + m\angle DNF$,

8. Name the property to justify the statement in question 7.

Substitution Property of Equality

Complete the proof.

	Statement	Reason
9.	$m\angle MNP = m\angle DFG$	Given
10.	$\angle MNP \cong \angle DFG$	Definition of Congruence